

On the uniqueness of flat thermodynamic geometries

Juan Martin N. Rodrigo* and Michael Francis Ian G. Vega II

National Institute of Physics, University of the Philippines Diliman, Quezon City 1101, Philippines

*Corresponding author: jrodrigo@nip.upd.edu.ph

Abstract

We investigate whether the ideal gas is the unique fluid equation of state corresponding to a flat thermodynamic geometry. Tentatively, we argue that the answer is no, and support this argument by exhibiting a choice of five virial coefficients that yield a thermodynamic curvature scalar which vanishes to fifth order in the number density. Furthermore, we see no obstructions in practice to finding coefficients that yield a curvature scalar which vanishes to all orders in the number density.

Keywords: thermodynamic geometry, statistical mechanics

1 Introduction

There has been much recent interest in the applications of Ruppeiner geometry, which is a theory imbuing thermodynamic state space with a metric [1]. This has been used to construct a partitioning line in the supercritical region of fluids, to compute fluid phase boundaries, and even to probe the nature of microscopic interactions in black holes, in many cases showing promising agreement with experiment [2–4]. (We recommend the paper by Ruppeiner [5] for a brief, readable, and fairly recent account of the theory and its predictions.)

To get started, one provides a Helmholtz free energy for the system, which suffices to describe all of its thermodynamic properties [6]. This allows for computation of the corresponding Ruppeiner metric, and from here one can examine geodesics and curvature. In this work, we will focus on a conjecture by Ruppeiner, who proposed that the sign of the thermodynamic scalar curvature R indicates the nature of microscopic interactions within the fluid, with $R > 0$ indicating predominantly repulsive interactions, and $R < 0$ indicating predominantly attractive interactions. Finally, vanishing curvature $R = 0$ indicates the absence of interactions [5]. This is consistent with the fact that the ideal gas corresponds to a metric with vanishing scalar curvature; that is, the ideal gas yields a flat metric [1].

Notably, to our knowledge, there has been no investigation into whether the ideal gas is the *unique* equation of state yielding a flat geometry. If this is the case, then the ideal gas is privileged not only physically (as the equation of state corresponding to the absence of interactions), but also geometrically. Furthermore, this provides us with an opportunity to probe Ruppeiner’s conjecture – if there exist non-ideal gas free energies with vanishing scalar curvature, do these necessarily correspond to nonzero interactions between the fluid particles? These are the questions we would like to address in this paper.

2 Flatness in thermodynamic geometry

First we describe the basic elements of Ruppeiner geometry; a much more thorough exposition can be found in Reference [1]. The metric is given by

$$dl^2 = \frac{1}{T} \left(-\frac{\partial^2 F}{\partial T^2} dT^2 + \frac{\partial^2 F}{\partial \rho^2} d\rho^2 \right), \quad (1)$$

where F , T , and $\rho \equiv N/V$ are the Helmholtz free energy, temperature, and number density of the fluid, respectively. Thus, given F , one can immediately compute the corresponding Ruppeiner metric, and from here obtain the Christoffel symbols, Riemann curvature tensor, and so on. The standard formula for the Ricci scalar, together with the expression in Eq. (1), gives

$$R = \frac{1}{\sqrt{g}} \left\{ \frac{\partial}{\partial T} \left[\frac{1}{\sqrt{g}} \frac{\partial}{\partial T} \left(\frac{1}{T} \frac{\partial^2 F}{\partial \rho^2} \right) \right] - \frac{\partial}{\partial \rho} \left[\frac{1}{\sqrt{g}} \frac{\partial}{\partial \rho} \left(\frac{1}{T} \frac{\partial^2 F}{\partial T^2} \right) \right] \right\}, \quad (2)$$

where g is the determinant of the metric, $g \equiv -(\partial^2 F / \partial T^2)(\partial^2 F / \partial \rho^2) / T^2$. We then set $R = 0$. Any solution $F(T, \rho)$ to this third-order nonlinear partial differential equation yields a metric with vanishing curvature, i.e. a flat metric. One can verify that the Helmholtz free energy for an ideal gas

$$F_{\text{ideal}}(T, \rho) = NkT \left[\ln \rho - \frac{3}{2} \ln(\gamma T) - 1 \right] \quad (3)$$

is a solution, where γ is a constant whose value is not relevant here, as it gets killed by the derivatives in Eq. (1). Finally, to ease notation, we'll write $f \equiv F$, $x \equiv T$, and $y \equiv \rho$ for the remainder of this paper.

3 An exact solution

Since we're looking to solve $R = 0$, it suffices to find where the numerator on the right-hand side of Eq. (2) vanishes. The corresponding equation is

$$\begin{aligned} 0 = & x^2 \left(\frac{\partial^2 f}{\partial x^2} \right) \left[\left(\frac{\partial^3 f}{\partial y^3} \right) \left(\frac{\partial^3 f}{\partial x^2 \partial y} \right) - \left(\frac{\partial^3 f}{\partial x \partial y^2} \right)^2 \right] \\ & - x \left(\frac{\partial^2 f}{\partial y^2} \right) \left\{ \frac{\partial^3 f}{\partial x \partial y^2} \left[\frac{\partial^2 f}{\partial x^2} + x \left(\frac{\partial^3 f}{\partial x^3} \right) \right] - x \left(\frac{\partial^3 f}{\partial x^2 \partial y} \right)^2 \right\} \\ & + \left(\frac{\partial^2 f}{\partial y^2} \right)^2 \left[2 \left(\frac{\partial^2 f}{\partial x^2} \right) + x \left(\frac{\partial^3 f}{\partial x^3} \right) \right]. \end{aligned} \quad (4)$$

A bit of squinting reveals that every term on the right-hand side is multiplied by a derivative ($\partial^2 f / \partial x^2$) or higher order, so $\partial^2 f / \partial x^2 = 0 \implies R = 0$. In particular, any free energy of the form

$$f(x, y) = xA(y) + B(y) \quad (5)$$

results in a vanishing curvature scalar. However, the resulting metric is degenerate, i.e. not a metric at all; a brief comparison against Eq. (1) reveals that the metric component g_{xx} is everywhere zero, so the metric determinant is always zero. Thus, we need to try something else. Motivated by the form of the free energy for the van der Waals gas (which we do not reproduce here, for reasons of space), we try

$$f(x, y) = xy[P(x) + Q(y)], \quad (6)$$

with P, Q free functions, and substitute this into Eq. (4). This leads to the ordinary differential equation

$$[2P'(x) + xP''(x)]^2 (2Q'(y) + 4yQ''(y) + y^2Q^{(3)}(y)) = 0. \quad (7)$$

There are two expressions here that we can independently set to zero in order to obtain a solution. Setting the expression in brackets to zero leads to a free energy of the form (5), so this solution is inadmissible. On the other hand, requiring that the expression in parentheses vanish and choosing the integration constants and free functions appropriately, we obtain the ideal gas free energy with an additional term:

$$f(x, y) = kxy \left(\ln y - \frac{3}{2} \ln(\gamma x) - 1 \right) + \lambda x + p(x)y, \quad (8)$$

where λ is a constant and p is a free function. However, although this is an exact solution to $R = 0$, we find it difficult to relate this to physical considerations. For instance, the corresponding entropy must obey certain constraints on concavity and extensivity [6]. Thus, we will try a different approach, which involves an approximation but is better physically motivated.

4 Flat free energies with nonzero virial coefficients

We look to the virial expansion for a gas, which expresses the fluid equation of state as a power expansion in the number density y . This is typically only valid in the low-density regime, where y is small. One can show that the Helmholtz free energy can be written in terms of the virial coefficients [6]

$$f(x, y) = kxy \left[\ln y - \frac{3}{2} \ln(\gamma x) - 1 + B_2(x)y + \frac{1}{2}B_3(x)y^2 + \frac{1}{3}B_4(x)y^3 + \dots \right]. \quad (9)$$

Since the additional terms in this series pick up extra powers of y , each term is less significant than the last. Thus, it is often reasonable to truncate this expansion after some chosen virial coefficient. Following a similar line of reasoning, we propose to find a free energy $f(x, y)$ such that the curvature scalar R vanishes up to a given order in y . Substituting Eq. (9) into Eq. (2), equating the scalar curvature to zero,

expanding in powers of y , and then retaining terms up to first order in the number density, we get

$$0 = \frac{1}{3} [x^2 B_2''(x) - 2x B_2'(x) - 3B_2(x)] + \frac{1}{9} y \left[-20x^2 B_2'(x)^2 + 24x B_2(x) B_2'(x) + 4x^4 B_2^{(3)}(x) B_2'(x) + 8x^3 B_2'(x) B_2''(x) + 36B_2(x)^2 + 3x^2 B_2''(x) - 12x B_2'(x) - 27B_3(x) \right] + \mathcal{O}(y^2). \quad (10)$$

The second order in y brings in the fourth virial coefficient B_4 , the third order brings in B_5 , and so on.

We now demand that the coefficients at each order vanish, starting with the zeroth-order coefficient. The corresponding equation is a linear second-order ordinary differential equation for B_2 , as can be verified by inspection, and it is readily shown that the general solution is

$$B_2(x) = c_1 x^{(3-\sqrt{21})/2} + c_2 x^{(3+\sqrt{21})/2}, \quad (11)$$

where c_1 and c_2 are constants of integration. Observe that this is the sum of one exponentially growing and one exponentially decaying term. If both integration constants are nonzero, then the second virial coefficient blows up (to positive or negative infinity, depending on the signs of the constants) in both temperature limits $x \rightarrow 0$ and $x \rightarrow \infty$.

Next, substituting this solution into the first-order coefficient in Eq. (10) and solving for B_3 (another linear second-order ordinary differential equation) yields

$$B_3(x) = c_3 x^{(5-\sqrt{61})/2} + c_4 x^{(5+\sqrt{61})/2} + A(x) x^{3-\sqrt{21}}, \quad (12)$$

where $A(x)$ is a complicated function of temperature dependent on c_1 and c_2 ; we do not reproduce it here. Together, Eqs. (11) and (12) represent a choice of virial coefficients that cause the curvature scalar to vanish to first order in the number density; it remains to specify the four integration constants in these two expressions.

In search of the simplest possible solution to Eq. (10), we put $c_1 = -\alpha$ and $c_2 = 0$ in Eq. (11), where α is a positive constant, and $c_3 = c_4 = 0$ in Eq. (12). A bit of simplifying leads to

$$B_2(x) = -\alpha x^{(3-\sqrt{21})/2}, \quad B_3(x) = \frac{4(-198 + 43\sqrt{21})}{57 - 12\sqrt{21}} \alpha^2 x^{3-\sqrt{21}}. \quad (13)$$

The choice $c_1 < 0$ and $c_2 = 0$ ensures that the second virial coefficient B_2 blows up to negative infinity as $x \rightarrow 0$, which is broadly consistent with predictions for the virial coefficients obtained from simple models. For instance, the van der Waals equation of state predicts $B_2(x) = a - b/x$ for positive constants a and b . Meanwhile, the choice $c_3 = c_4 = 0$ is made for convenience, because any choice for c_3 and c_4 yields a mathematically valid solution to Eq. (10). Finally, we note that the coefficients in Eq. (13) have the same limiting behavior: the numerical factor in front of B_3 is negative, so that it also blows up to negative infinity as $x \rightarrow 0$, and both virial coefficients tend to zero in the $x \rightarrow \infty$ limit.

We found no barriers in practice to requiring that Eq. (10) hold up to higher orders in y ; this only required an appropriate choice of integration constants at each new order. An example is shown in Fig. 1, where we've chosen five virial coefficients that yield an R that vanishes up to $\mathcal{O}(y^5)$. In fact, we conjecture that it is possible to choose virial coefficients such that the scalar curvature vanishes to *all* orders of y . Preliminary work with Mathematica suggests that requiring the scalar curvature to vanish up to $\mathcal{O}(y^k)$ only requires solving a linear second-order ordinary differential equation in B_{k+1} . We hope to find a more compelling argument in support of this observation.

4.1 Virial coefficients from microscopic interactions, and the inverse problem

The coefficients in Eq. (9) are also significant from a statistical mechanics viewpoint; these can be derived by accounting for interactions between the constituent gas particles. What, then, are the microscopic interactions leading to the solutions in Eq. (13)? As a start, we look at the second virial coefficient in terms of the intermolecular potential energy $u(r)$, which is given by [7]

$$B_2(x) = -2\pi \int_0^\infty r^2 \left(e^{-u(r)/kx} - 1 \right) dr. \quad (14)$$

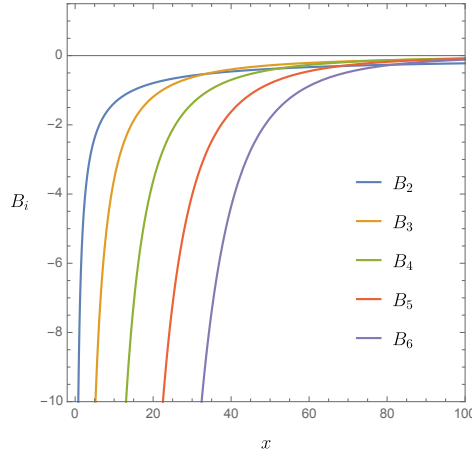


Figure 1: A choice of five virial coefficients that yield a thermodynamic scalar curvature that vanishes up to fifth order in the number density y . Each of these coefficients blows up exponentially to negative values as the temperature goes to zero. (Strictly speaking, the virial coefficients have differing units; we display them all on the same plot to get a qualitative understanding of their behavior.)

Equating the second virial coefficient from Eq. (13) to the expression here yields an equation for $u(r)$. Thus, while Eq. (14) is typically provided as a way to compute B_2 from a known interaction, here we consider the inverse problem of determining the interaction from the virial coefficient. We argue that it is easier to interpret the intermolecular potential, and therefore answer questions about the physicality (or unphysicality) of the solutions we have found so far, than it is to examine the bare virial coefficients. However, the form of this equation is unfamiliar to us, and it is unclear how to proceed in solving for $u(r)$. This is something we plan to investigate further.

5 Conclusions

We have discovered a choice of five virial coefficients that yield a thermodynamic scalar curvature which vanishes up to fifth order in the number density. The corresponding gas is manifestly non-ideal, but due to this vanishing scalar curvature, Ruppeiner geometry predicts that effective interactions between the fluid's constituent particles are negligible. We found no barriers in practice to continuing this construction to higher orders; indeed, we conjecture that it is possible to choose virial coefficients such that the scalar curvature vanishes to *all* orders in y . It remains to be shown which microscopic interactions lead to the virial coefficients discovered here.

Finally, we point out that the Euclidean metric is privileged as being flat in differential geometry, since all other flat metrics can be related to it via an appropriate coordinate transformation. It remains an open question what set of transformations in the free energy relates the ideal gas and the flat free energies discovered in this paper. Presumably, these transformations are related to coordinate transformations in thermodynamic geometry. We leave further investigation of these concerns to future work.

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